Diophantine Equations: The Answer Really Is 42

Josephine Lee

1 Introduction: Sum of Cubes Question

Deep Thought's conclusion found in Douglas Adams' *The Hitchhiker's Guide to the Galaxy* - that "the answer to the great question... of life, the universe and everything... is... forty-two" — was paradoxically reflected in completing the final piece of the puzzle of a question mathematicians have been ruminating on since 1954. The question was formulated in Oxford University in 1954 by Professor Louis Joel Mordell: what integer solutions are there to the equation

$$x^3 + y^3 + z^3 = k,$$

where k is less than or equal to 100?

2 History

Answers were most quickly discovered for small cases:

By Fermat's Last Theorem, for $k = 0$, only trivial solutions $(a^3 + (-a)^3 + 0^3 = 0)$ exist.

Similarly, for $k = 1$ and $k = 2$, infinite solutions exist:

$$(9b^4)^3 + (3b - 9b^4)^3 + (1 - 9b^3)^3 = 1$$

(discovered by K. Mahler in 1936)

$$(1 + 6c^3)^3 + (1 - 6c^3)^3 + (-6c^2)^3 = 2$$

(discovered by A.S. Verebrusov in 1908)

Over several years, using sophisticated computers, findings were computed for nearly every $k < 100$. We can walk through the algorithm used to find the solutions for $k = 39$ in 1993:

If $k = 3$ then $k \equiv 3 \bmod 9$, so $x \equiv y \equiv z \equiv 1 \bmod 3$. If x, y, z are the same sign, $x = y = z = 1$. Otherwise, assume that x, y have the same sign and z have the opposite sign, which makes $|x + y| \geq |z| \geq 1$. We can solve $z^3 \equiv 3 \bmod 3$ such that $n = x + y$. Since z and n have different signs, $1 \leq |z| \leq |n|$. From the findings in 1989 by D. R. Heath-Brown, we know that $(n, 3) = 1$, and that $n = a^3 + 3b^3 + 9c^3 - 9abc$ such that a, b, c are integers and

$$z \equiv (3c^2 - ab)(b^2 - ac)^{-1} \bmod n,$$

where $(b^2 - ac, n) = 1$, and z and n have different signs. Then, we find the solutions to $x + y = n$ and $x^3 + y^3 + z^3 = n$ to find unique values for z, yielding:

$$x = \frac{n+d}{2}, y = \frac{n-d}{2}, d = \sqrt{D}, \text{ and } D = \frac{1}{3}[4(\frac{3-z^3}{n} - n^2)]$$

where D is a square root of an integer.

In 1993, researchers used the Cyber 205 vector computer to run a vectorized version of the Euclidean algorithm. Arithmetic operations were able to be vectorized, and the algorithm became capable of solving the equation $w_i x_i \equiv 1 \bmod n$, i = 1, 2, ..., at vector speed. It was through this algorithm that the first solution for $k = 39$ was found: (134476, 117367, -159380).

Throughout the decade, the unsolved values of $k \leq 100$ eventually decreased to 33 and 42.

3 Finding the Answers for $k = 42$ and $k = 33$

In 2019, Andrew Brooker from the University of Bristol used a supercomputer to finally find the solution for 33, but the solution for 42 remained elusive. Brooker decided to ask for help from Professor Sutherland from MIT, who used Charity Engine to find the answer for $x^3 + y^3 + z^3 = 42$. Charity Engine utilizes more than 500,000 home PCs with unused computer power, and it still took over a million hours to find the solution. The final successful run took a few weeks.

Employing otherwise unused computational resources makes the Charity Engine network more energy-efficient. The Charity Engine takes advantage of the large portion of the CPU that would otherwise go to waste, making the amount of electricity needed to conduct the computations much less than what would be needed for a supercomputer to do the same work. Brooker and Sutherland also applied several adaptations and optimized the code to make it run more efficiently for a gigantic distributed computation. In contrast, a supercomputer would have to make a long computation run.

The core of the algorithm the Charity Engine uses is as follows:

Suppose that (x, y, z) is a solution to $k = x^3 + y^3 + z^3$ and WLOG that $|z| \leq |y| \leq |x|$. Then:

$$kz^3 = x^3 + y^3 = (x + y)(x^2 - xy + y^2).$$

When $k - x^3 = 0, y = -x$, leading to every value of x being a solution.

We can set $d = |x + y| = |x| + y$ sgnx, where

$$sgn(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Thus, d divides $|k - z^3|$:

$$\frac{|k - z^3|}{z} = x^2 + xy + y^2 = |x|(2|x| - d) + (d - |x|)^2 = 3x^2 - 3d|x| + d^2$$

so that the solutions to x and y are

$$\frac{1}{2}\text{sgn}(k-z^3)(d \pm \sqrt{\frac{4|k-z^3|-d^3}{3d}})$$

So, when we run through all the divisors of $|k-z^3|$, we can find all corresponding values of x and y with $\min(|x|,|y|,|z|) \leq B$ in time $O(B^{1+\epsilon})$.

With the solution for $k = 33$, the search bound B was 10^{16}, but for 42, B had to equal 10^{17}, making it much harder for a supercomputer to solve.

4 What Is Still Uncovered in the Sum of Cubes Problem

The Charity Engine found solutions for k =42, 165, and 906:

$$(-80538738812075974)^3 + 80435758145817515^3 + 12602123297335631^3 = 42,$$

$$(-385495523231271884)^3 + 383344975542639445^3 + 98422560467622814^3 = 165, \text{and}$$

$$(-74924259395610397)^3 + 72054089679353378^3 + 35961979615356503^3 = 906$$

The only remaining unsolved cases up to 1,000 are 114, 390, 579, 627, 633, 732, 921 and 975.

5 Other Diophantine Equations

5.1 The Hardy-Ramanujan Equation

The equation that all taxicab numbers satisfy is

$$w^3 + x^3 = y^3 + z^3$$

where w, x, y, z are all integers.

Here are all the known taxicab numbers:

$$2 = 1^3 + 1^3$$

$$1729 = 1^3 + 12^3 = 9^3 + 10^3$$

$$87,539,319 = 167^3 + 436^3 = 228^3 + 423^3 = 255^3 + 414^3$$

$$6,963,472,309,248 = 2421^3 + 19083^3 = 5436^3 + 18948^3 = 10200^3 + 18072^3 = 13322^3 + 16630^3$$

$$48988659276962496 = 38787^3 + 365757^3 = 107839^3 + 362753^3 = 205292^3 + 342952^3$$

$$24,153,319,581,254,312,065,344 = 582162^3 + 28906206^3 = 3064173^3 + 28894803^3$$

5.2 The Erdős–Straus Conjecture

The Erdős–Straus Conjecture states that for every integer $n \geq 2$, there exists positive integers x, y, and z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

The conjecture has yet to be proved, as of 2019. The most progress we have is from Mordell in 1967, that a counterexample must be one of six infinite arithmetic progressions modulo 840.

6 Applications of Diophantine Equations

Let's look at a few applications of Diophantine equations:

6.1 Solar System

When investigating the stability of the solar system, the near resonance of the frequencies of planets are associated with denominators.

The dynamical properties of a rotation of angle α on a circle of length L are radically different depending on whether the parameter $\frac{\alpha}{L}$ is rational or not. If $\frac{\alpha}{L}$ is rational, the orbits are all periodic; if not, they are all dense.

Carl L. Siegal, a German mathematician who specialized in analytic number theory, discovered a suitable diophantine inequality on the frequencies to ensure that they are not too close to resonance. His theorem states as follows:

All normalized f with multiplier λ in a subset of full measure of the circle have a guaranteed radius of linearization. The bound is dependent only on the diophantineness order of λ. A diophantine number θ of order $v + 1$ has the property

$$|\theta - \frac{m}{n}| > \frac{Q}{n^{v+1}}$$

where m is the nearest integer to $n\theta$, and $|\lambda^n - 1| = \Omega(\log n)$ as $n \longrightarrow \infty$.

6.2 Computational Complexity Theory

Another application of Diophantine equations lies in computational complexity theory. Matiyasevich's theorem states that a set is recursively enumerable if and only if it is also Diophantine. Thus, the complexity class of recursively enumerable problems are also associated with Diophantine equations, such as the Halting Problem:

Given a computer program and an input, will the program terminate or will it run forever?

The halting set that represents the Halting Problem is

$$K = (i, x) \mid \text{program } i \text{ halts when run on input } x$$

As stated before, the halting set is recursively enumerable; in other words, there exists a computable function that lists all the pairs (i, x) it contains.

Sadly, as proved by Alan Turing in 1936, a general algorithm solution to the Halting Problem does not exist. The term "Turing degree" is named after him, which refers to a measure of the noncomputability in any solution.

7 Conclusion

The Sum of Cubes question took nearly 70 years to be solved. Interestingly, sometimes making the computer more powerful doesn't provide the answer. There are still many discoveries to be made when k is increased and in other types of Diophantine equations. Diophantine equations have a long history and remain important in modern applications in astronomy and computer science.

References

"A Brief but Historic Article of Siegel." Research Gate, n.d. https://www.researchgate.net/publication/267224695$_{AB}rief_{b}ut_{H}istoric_{A}rticle_{o}f_{S}iegel.$

Booker, Andrew R. "Cracking the Problem with 33." July 19, 2019. https://link.springer.com/article/10.1007/s40993-019-0162-1

Booker, Andrew R., and Andrew V. Sutherland. "On a Question of Mordell." arXiv.org, July 9, 2020. https://arxiv.org/abs/2007.01209.

Heath-Brown, Roger., Lioen, Walter M., and Te Riele, Herman. "On Solving the Diophantine Equation on a Vector Computer." Research Gate, n.d. https://www.researchgate.net/publication/265428623$_{O}n_{s}olving_{t}he_{D}iophantine_{e}quation_{o}n_{a}{}_{v}ector_{c}omputer.$

Staff, Science X. "Sum of Three Cubes for 42 Finally Solved-Using Real Life Planetary Computer." Phys.org. September 6, 2019. https://phys.org/news/2019-09-sum-cubes-solvedusing-real-life.html.

"Sutherland Helps Solve Decades-Old Sum-of-Three-Cubes Puzzle." MIT School of Science. Accessed Jan uary 26, 2021. https://science.mit.edu/sutherland-helps-solve-decades-old-sum-of-three-cubes-puzzle/.

Veisdal, Jørgen. "Famous Diophantine Equations." Medium. Cantor's Paradise, September 22, 2019. https://medium.com/cantors-paradise/famous-diophantine-equations-84073467d366.

"What Are the Applications of Diophantine Equations." Research Gate, n.d. https://www.researchgate.net/post/What-are-the-applications-of-diophantine-equations-where-do-we-apply-the-higher-degree-diophantine-equations.

"Why Is Turing's Halting Problem Unsolvable?" Scientific American. Scientific American, February 19, 2006. https://www.scientificamerican.com/article/why-is-turings-halting-pr/.